FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Main)-2020

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. D

Sol. $T_1 = \frac{2\pi R}{v_0}$

$T_2 = \sqrt{\frac{2h}{g}}$

$N = \sqrt{\frac{2h}{g}} \times \frac{v_0}{2\pi R}$

2. D

Sol. $Kx = \frac{N}{\sqrt{2}} = mg$

$x = \frac{mg}{K}$

3. B

Sol. Let *x* be the depth of point *P* from surface

App. depth of point *P* from surface = $\frac{x}{\mu}$

App. depth of image of P from surface = $\frac{x+2h}{\mu}$

So, separation between two = $\frac{x+2h}{\mu} - \frac{x}{\mu}$

$\Rightarrow \frac{2h}{\mu}$

4. B

Sol. $\Delta Q = \Delta U + W$

$nCdT = nCvdT + PdV$

$C = C_V + P\frac{dV}{dT}$ …(1)

$P = 2P_0 + \frac{2P_0}{V_0}(V - V_0)$

$= 2P_0 + \frac{2P_0V}{V_0} - 2P_0$

$P = \frac{2P_0V}{V_0}$

$PV = RT$

$\frac{RT}{V} = \frac{2P_0V}{V_0}$

$RT = \frac{2P_0}{V_0}V^2$

$\frac{dT}{T} = \frac{2dV}{V}$

$\frac{dV}{dT} = \frac{V}{2T}$ …(2)

from (1) and (2)

$C = \frac{3R}{2} + \frac{nRT}{V} \times \frac{V}{2T}$

$\Rightarrow \frac{3R}{2} + \frac{R}{2} \Rightarrow 2R$

5. C

Sol. Resistance across capacitor,

$R = \frac{4 \times 8}{4+8} = \frac{32}{12}\Omega$

$\tau_C = CR = 16\ \mu$ sec

$i = i_0 e^{-t/\tau_c} = \left(\frac{160/6}{32/12}\right) e^{-t/\tau_c} \quad \Rightarrow 10\, e^{-t/\tau_c}$

After $t = \tau_c$, $i = \frac{10}{e}A$

This is distributed in inverse ratio of resistance,

$i_{1\Omega} = \frac{2}{3}\left(\frac{10}{e}\right) = \frac{20}{3e}$ A

6. A

Sol. $Mg = 2T\cos\theta$

$T = mg$

7. C

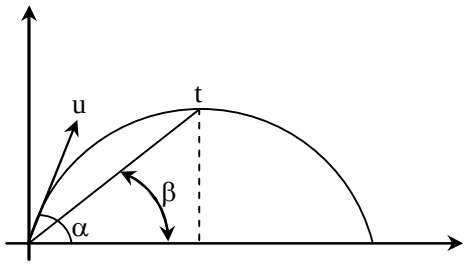
Sol. $P_{KE} = \frac{1}{2}m(2v)^2 = 2mvR$

KE of S = $\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \Rightarrow \frac{3}{4}mv^2$

$\frac{(KE_P)}{(KE_S)} = \frac{8}{3}$

8. B

Sol. $\tan\beta = g/x = \frac{\mu\sin\alpha t - 1/2gt^2}{\mu\cos\alpha t} = \tan\beta$



$\sin\alpha - \tan\beta.u\cos\alpha = gt/2$

$u = \frac{gt\cos\beta}{2\sin(\alpha-\beta)}$

9. B

Sol. $Kx = 2g$; $x = \frac{2g}{k} = \frac{1}{2}$

$mgx = \frac{1}{2}kx^2 + \frac{1}{2}mv^2$

$v = \sqrt{2gx - \frac{kx^2}{m}} = 2\sqrt{2}$

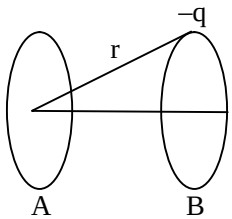
10. B

Sol. $\int \Delta P = \int F dt$

$P_2 - P_1 \Rightarrow F\int_0^t dt$

11. C

Sol. $V_A = V_1 + V_2$



$V_A = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{R} - \frac{1}{\sqrt{R^2+a^2}}\right)$

$V_B = \frac{q}{4\pi\varepsilon_0}\left[-\frac{1}{R} + \frac{q}{r}\right]$

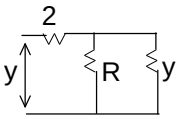
$$V_A - V_B = \frac{2q}{4\pi\varepsilon_0}\left(\frac{1}{R} - \frac{1}{\sqrt{R^2 + a^2}}\right)$$

12. C

Sol. $B = \frac{\mu_0}{4\pi}\frac{1^0}{d}(\sin\alpha + \sin\beta)$

13. C

Sol.



$$y = \frac{Ry}{R+y} + 2R$$

& $x = \frac{Ry}{R+y}$

14. B

Sol. $\vec{v}_{1m} = -m^2\ \vec{v}_{0m}$ (along x-axis)

$\vec{v}_{1m} = m\,\vec{v}_{0m}$ (along y-axis)

The Range of projective will be

Range = $\frac{v^2 \sin 2\theta}{g} = \frac{(gR)\sin 90^\circ}{g} = R$

So particle will at 3f from pole of mirror with velocity $\frac{-v}{\sqrt{2}}\hat{i} - \frac{v}{\sqrt{2}}\hat{j}$

15. A

Sol. Here the string is fixed at both the ends so equation of standing wave as a function of distance is given, y = A sin x

where A is maximum amplitude

2 = 4 sin kx & kx = $\frac{\pi}{6}$ and $\frac{5\pi}{6}$

Where k = $\frac{2\pi}{\lambda}$ & λ = 300 cm.

$\frac{2\pi}{\lambda}x_1 = \frac{\pi}{6} \Rightarrow \frac{2\pi}{300}x_1 = \frac{\pi}{6} \Rightarrow x_1 = 25$ cm

$\frac{2\pi}{\lambda}x_2 = \frac{5\pi}{6} \Rightarrow \frac{2\pi}{300}x_2 = \frac{5\pi}{6} \Rightarrow x_2 = 125$ cm

So, separation between two points = 125 – 25

= 100 cm

16. D

Sol. $I = \frac{dq}{dt}$

q = it + a

$V = \frac{q}{C}$

$V = \frac{it + a}{C}$

$\therefore$ V is proportional to time.

17. C

Sol. $\vec{V} \perp \vec{B}$ Therefore path of particle is a circle. In magnetic field speed to particle of remains constant. Therefore distance moved by particle in time $t = \frac{\pi}{B_0\alpha}$ is v_0t or $\frac{\pi v_0}{B_0\alpha}$ and magnitude of velocity is always v_0.

18. A

Sol. $y = f(ax \pm bt)$ is general wave equation

At t = 0, $y = \frac{1}{\sqrt{1+x^2}}$

At t = 1, $y = \frac{1}{\sqrt{2-2x+x^2}} = \frac{1}{\sqrt{1+(x-1)^2}}$

Here a = 1 and b = 1

wave velocity, $v = \frac{b}{a}$ = 1 m/sec towards +ve x.

19. C

Sol. $4[1 + 11 \times 10^{-6}\,\Delta\theta] = 3.992[1 + 19 \times 10^{-6}\Delta\theta]$

$\Delta\theta = 250^oC$

So $\theta = 250 + 30 = 280^oC$

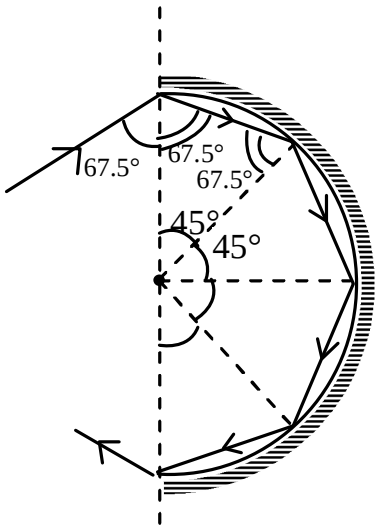
20. B

Sol. Excluding at ∞ and $-\infty$, one maxima will be observed at symmetrical position and one each over both this central one.

SECTION – B

21. 5

Sol.



22. 5

Sol. Velocity of centre of mass of cylinder w.r.t. plank, $v_{cp} = 20 - 10 = 10$ m/s

We have $v_{cp} = R\omega \Rightarrow \omega = \frac{10}{R}$

Kinetic energy KE = $\frac{1}{2}mv_{cm}^2 + \frac{1}{2}I\omega^2$

$= \frac{1}{2} \times 2 \times 400 + \frac{1}{2} \times 2 \times \frac{R^2}{2} \times \frac{100}{R^2}$

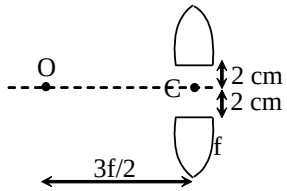
= 450 J

i.e. $\boxed{n = 5}$

SECTION – C

23. 00006.00

Sol.



For upper part of lens

$u = -\frac{3f}{2}$, $h_0 = 2$ cm

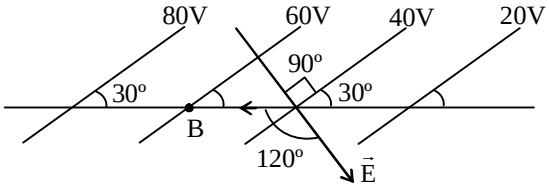
$m = \frac{h_i}{h_0} = \frac{f}{f + \left(\frac{-3f}{2}\right)} = 0.5$

$h_i = 2 \times 0.5 = 1$ cm

i.e. image is formed at a height of 3 cm (i.e. 2 + 1) from main principle axis. (above principle axis)
Similarly for lower part, image is formed 3 cm below main principle axis. Hence distance between image = 3 + 3 = 6 cm.

24. 00400.00

Sol. Electric field lines are perpendicular to equipotential surfaces and electric potential decreases along electric field.



From A $\rightarrow$ B

| dV | = –E dr cos 120°

$20 = E \times (10 \times 10^{-2}) \times \frac{1}{2}$

E = 400 N/C

25. 00100.00

Sol. Let, ${}_{Z}^{A}X \rightarrow {}_{A-2}^{A-4}Y + {}_{2}^{4}He$

$K_\alpha = \frac{m_y}{m_y + m_\alpha} Q = \frac{A-4}{A} Q$

$48 = \frac{A-4}{A} \times 50 \Rightarrow A = 100$

Chemistry

PART – II

SECTION – A

26. D

Sol. $pH = -\log\left[H^+\right]$ and $\left[H^+\right] = \left[H_3O^+\right]$

27. A

Sol. in exothermic reaction $\Delta H < 0$

$$\ln K_{eq} = \ln K_0 - \frac{\Delta H}{RT}$$

28. C

Sol. for ideal gas 'v' at given T and P should be 24.63 L

29. D

Sol. m = - ℓ to ℓ

30. C

Sol. $\frac{1}{\lambda} = Rz^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \Rightarrow \frac{1}{\lambda} = Rz^2\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$

31. B

Sol. no. of equivalent of substance deposite or liberated = change passed in faraday
Since 10% Cd by mass with 4.5 gm of Hg i.e. 90%

$$\therefore 10\% = \frac{4.5}{90} \times 10 = 0.5 \text{ gm Cd}$$

32. C

Sol. $(n-1)K.t = \left(\frac{1}{A_t^{n-1}} - \frac{1}{A_0^{n-1}}\right)$

33. C

Sol. $\Delta S = n.c.p.\ln\frac{T_2}{T_1}$

34. C

Sol. F_{CC} has 0.26 as void fraction

35. B

Sol. $\theta = 180^o$

36. A

Sol. no. of e^- in $NO_3^- = 32$ = no. of e^- in CO_3^{2-}

37. A

Sol. inactive due to pos

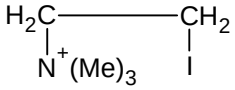
38. B

Sol. Al act as R.A

39. C
Sol. EAN = [atomic no. – O.S + 2 × L.P]

40. A
Sol. properties based

41. A
Sol. anti form is most stable



42. D
Sol. acidic strength $\propto$ stability of C.B

43. A
Sol. stability $\propto$ no. of $\propto$ - H

44. B
Sol. product having chiral centre

45. B
Sol. explanation ozonolysis

SECTION – B

46. 8
Sol. as (n – 1) = 7

47. 3
Sol. $x = \frac{3N}{2\times100} \therefore \frac{200x}{N} = 3$

SECTION – C

48. 00004.00
Sol. Here G.I generating system = 2, no. G.I. = $2^2 = 4$

49. 00003.00
Sol. $CH_3 - \overset{+}{C}H_2$
$\propto H = 3$

50. 00008.00
Sol. z = 3
w = 5 $\therefore$ z + w = 8

Mathematics

PART – III

SECTION – A

51. B

Sol. $\tan\left(3\times\frac{\pi}{18}\right)=\frac{3\tan\left(\frac{\pi}{18}\right)-\tan^3\left(\frac{\pi}{18}\right)}{1-3\tan^2\left(\frac{\pi}{18}\right)}$

52. C

Sol. Roots are $\cos\theta, \sin\theta, 1$

$\cos\theta=1 \quad \Rightarrow 0, 2\pi$

$\sin\theta=1 \quad \Rightarrow \frac{\pi}{2}$

$\sin\theta=\cos\theta \quad \Rightarrow \theta=45^0, 225^0$

53. A

Sol. $\left(x^2-y^2\right)\left(x^2+y^2\right)=\underbrace{3789108}_{\downarrow}$

$(x-y)(x+y)\left(x^2+y^2\right)$ even

For LHS to be even, x and y both must be, even or both must be odd.

$\Rightarrow (x-y), (x+y), \left(x^2+y^2\right)$ must be even integer.

$\Rightarrow (x-y)(x+y)\left(x^2+y^2\right)$ must be divisible by 8.

But RHS is not divisible by 8.

No solution.

54. A

Sol. $F\rightarrow(T\wedge F)\vee(F\wedge T)$

$F\rightarrow(F\vee F)$

$F\rightarrow F=T$

55. B

Sol. It is a concept.

56. A

Sol. $4\cos\theta\cos2\theta\cos3\theta=1$

$\Rightarrow 2(\cos3\theta+\cos\theta)\cos3\theta=1$

$\Rightarrow 2\cos^2 3\theta+2\cos3\theta\cos\theta=1$

$\Rightarrow 1+\cos6\theta+\cos4\theta+\cos2\theta=1$

$\Rightarrow \cos4\theta(2\cos2\theta+1)=0$

$\Rightarrow 4\theta=(2n+1)\frac{\pi}{2}$

or $2\theta=2n\pi+\frac{2\pi}{3}$

$\theta=(2n+1)\frac{\pi}{8} \qquad \theta=n\pi\pm\frac{\pi}{3}$

θ between 0 & π

$\theta = \frac{\pi}{8}, \frac{\pi}{3}, \frac{3\pi}{8}, \frac{2\pi}{3}, \frac{5\pi}{8}, \frac{7\pi}{8}$

57. D

Sol. $g(x) = -f(x)$

$\therefore h(x) = 0$ which is many one into.

58. B

Sol. $-\frac{\pi}{2} \le \sin^{-1}{}_{xi} \le \frac{\pi}{2}$

so only possible if

$x_1 = x_2 = x_3 = ... = x2n = 1$

$\sum_{i=1}^{2n} xi = 2n$

59. A

Sol. $\lim_{n\to\infty} \frac{\cos n!}{\left(n^{1-l}\right)\left(1+\frac{1}{n}\right)}$

$0 < k < 1$

$\Rightarrow k-1$ is negative

as $n \to \infty$

$-1 \le \cos n! \le 1 \Rightarrow$ finite

$n^{1-k} \to \infty$

$1+\frac{1}{n} \to 1$

so, $\lim_{n\to\infty} \frac{\cos n!}{n^{1-k}\left(1+\frac{1}{n}\right)}$

60. A

Sol. Replace $x \to x-3$

$f(x-3) = \sqrt{\sin^{-1}(x-3)} + \sqrt{\cos^{-1}(x-3)}$

$0 \le x-3 \le 1$

$3 \le x \le 4$

$xt[3, 4]$

61. C

Sol. $\sqrt{1-x^6} + \sqrt{1-y^3} = a^3\left(x^3 - y^3\right)$

put $x^3 = \sin\theta$ & $y^3 = \sin\theta$

62. D

Sol. $f(0) = 1, f(1)$

$f'(x) = 0$ must have at least one root in (0, 1)

$f'(x) = x^{m-1}m(x-1)^n + nx^m(x-1)^{n-1}$

$= x^{m-1}(x-1)^{n-1}\left[(m+n)x - n\right]$

$x = 0, 1, \dfrac{m}{m+n}$

63. A

Sol. $\dfrac{c^4}{r^2} = a^2\operatorname{cosec}2\theta + b^2\sec 2\theta$

$r^2 = \dfrac{c^2}{\left(a^2\operatorname{cosec}2\theta + b^2\sec 2\theta\right)_{min}}$

$= \dfrac{c^4}{(a+b)^2}$

$r_{max} = \dfrac{c^2}{a+b}$

64. B

Sol. $L = \prod_{n=3}^{\infty}\dfrac{n-2}{n}\times\dfrac{n+2}{n}$

$\lim\limits_{n\to\infty}\dfrac{1}{3}\times\dfrac{2}{4}\times\dfrac{3}{5}\times.....\times\dfrac{n-3}{n-1}\times\dfrac{n-2}{n}\times.....\dfrac{n+1}{n-1}\times\dfrac{n+2}{n}$

$\lim\limits_{n\to\infty}\dfrac{2\left(n^2+3n+2\right)}{I_2\left(n^2-n\right)} = \dfrac{1}{16}$

Similarly find M & N also

$L^{-1} = 6, M^{-1} = \dfrac{3}{2}, N^{-1} = \dfrac{1}{2}$

65. C

Sol. $\sin\left\{\dfrac{\pi}{900}\left[\sum_{r=1}^{10}\left(r^2+r+1\right)\right]\right\}$

$\Rightarrow \sin\left\{\dfrac{\pi}{900}\left[\dfrac{10\times11\times21}{6}+\dfrac{10\times11}{2}+10\right]\right\}$

$\sin\left(\dfrac{\pi}{900}\times450\right) = 1$

66. A

Sol. Both S_1 & S_2 are true.

67. C

Sol. Total ways of choosing 4 socks out of 12 pair (24 socks) is = $24\times23\times22\times21$

The number of ways in which on pair is seteated = $24\times22\times20\times18$

The prob. of not getting a pair.

$= \dfrac{24\times22\times20\times18}{24\times23\times22\times21} = \dfrac{120}{161}$

Hence prob. of getting at least me pair.

$1-\dfrac{120}{161} = \dfrac{41}{161}$

68. A

Sol. Property of determinant

69. B

Sol. Let $\vec{p} = a\hat{i}, \vec{q} = a\hat{j}, \vec{r} = a\hat{k}$

let $\vec{x} = x\hat{i} + y\hat{j} + \hat{k}$

Put in given equation

70. B

Sol. $\int \frac{dt}{t^2 + 2xt + x^2 - x^2 + 1}$

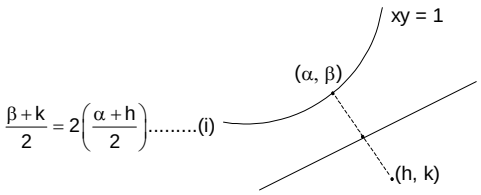
$\int \frac{dt}{(t+x)^2 - (x^2 - 1)} \qquad \therefore x^2 > 1$

$= \frac{1}{2\sqrt{x^2 - 1}} \log \left(\frac{t + x - \sqrt{x^2 - 1}}{t + x + \sqrt{x^2 - 1}} \right) + C$

SECTION – B

71. 7

Sol.



$\frac{\beta + k}{2} = 2\left(\frac{\alpha + h}{2}\right)$.....(i)

$\frac{k - \beta}{h - \alpha} = -\frac{1}{2}$ .....(ii)

from (i) & (ii)

$\alpha = \frac{4k - 3h}{5} \Big| \beta = \frac{4h + 3k}{5}$

$\alpha\beta = 1$ get curve.

72. 4

Sol. $\log_{10}^{\alpha} - \log_{a}^{10^{-4}}$

$\frac{\log_{10}^{a} + 4\log_{a}^{10}}{2} \geq (4)^{\frac{1}{2}}$

SECTION – C

73. 00002.50

Sol. $a - 1 = 0 \Rightarrow a = 1$

$$L = \lim_{x\to 0} \frac{ae^x + 2\sin 2x + b\cos bx}{2}$$

$$1 + b = 0 \Rightarrow b = -1$$

$$L = \lim_{x\to 0} \frac{ae^x + 4\cos 2x - b^2 \sin bx}{2} = \frac{a+4}{2} = \frac{5}{2} = 2.5$$

74. 00943.00

Sol. $^7P_4 \times 1 + \frac{^7P_3}{2} \times 1 - 2 = 943.$

75. 00002.07

Sol. $R = SH = 5$

sides are $5\sqrt{2}, 5\sqrt{2}, 10$

$$r = \frac{\Delta}{S}$$